

UniNotes

MATHEMATICS

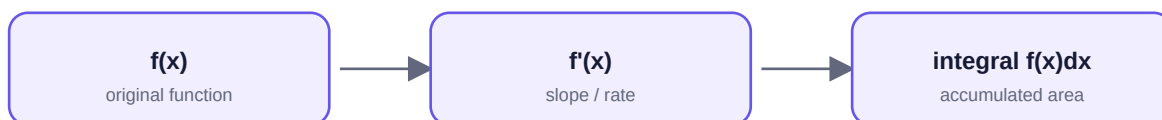
Calculus Fundamentals

Differentiation and integration rules, interpretation and fast revision examples.

How to use these notes

Start with the key definitions, then use the tables/diagrams to connect ideas. Finish with the exam checklist. The aim is high information density: short wording, clear concepts, no filler.

1. The big picture



Differentiation measures instantaneous rate of change (gradient). Integration measures accumulation/area and reverses differentiation up to a constant.

2. Differentiation rules

Power	$\frac{d}{dx}(x^n) = n x^{n-1}$
Constant	$\frac{d}{dx}(c) = 0$
Product	$(uv)' = u'v + uv'$
Quotient	$(\frac{u}{v})' = \frac{u'v - uv'}{v^2}$
Chain	$\frac{d}{dx} f(g(x)) = f'(g(x))g'(x)$

Example: $y = (3x^2 + 1)^5$. Chain rule: $y' = 5(3x^2+1)^4 \times 6x = 30x(3x^2+1)^4$.

3. Stationary points

Stationary points occur where $f'(x)=0$. Solve $f'(x)=0$, substitute x into $f(x)$, then classify. If $f''(x)>0$ it is a local minimum; if $f''(x)<0$ it is a local maximum; if $f''(x)=0$ further analysis is needed.

4. Integration rules

Power	$\int x^n dx = \frac{x^{n+1}}{n+1} + C, n \neq -1$
Constant multiple	$\int k f(x) dx = k \int f(x) dx$

Definite integral	$\text{integral } [a,b] f(x)dx = F(b)-F(a)$
$1/x$	$\text{integral } 1/x \, dx = \ln x + C$

Example: $\text{integral } (6x^2 - 4x + 3)dx = 2x^3 - 2x^2 + 3x + C$.

5. Area and fundamental theorem

A definite integral gives signed area: regions below the x-axis contribute negatively. Geometric area may require splitting the interval and taking absolute values.

Fast check

Differentiate your indefinite integral. If you do not recover the original integrand, the integration is wrong.

Exam-ready checklist

- Differentiate powers, products, quotients and composites.
- Find and classify stationary points.
- Include $+C$ for indefinite integrals.
- Use $F(b)-F(a)$ for definite integrals.
- Distinguish signed integral from total geometric area.

Fast recall

If you cannot explain a concept in one or two sentences without looking, revisit that section. Aim to recall the definition, mechanism and one example.

Uni Notes keeps revision focused: concise explanations, useful detail, and diagrams only where they improve understanding.